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Water or Glycol for Direct Liquid Cooling?

A quantitative tradeoff analysis across thermal, operational and programmatic domains

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THE COOLANT DECISION

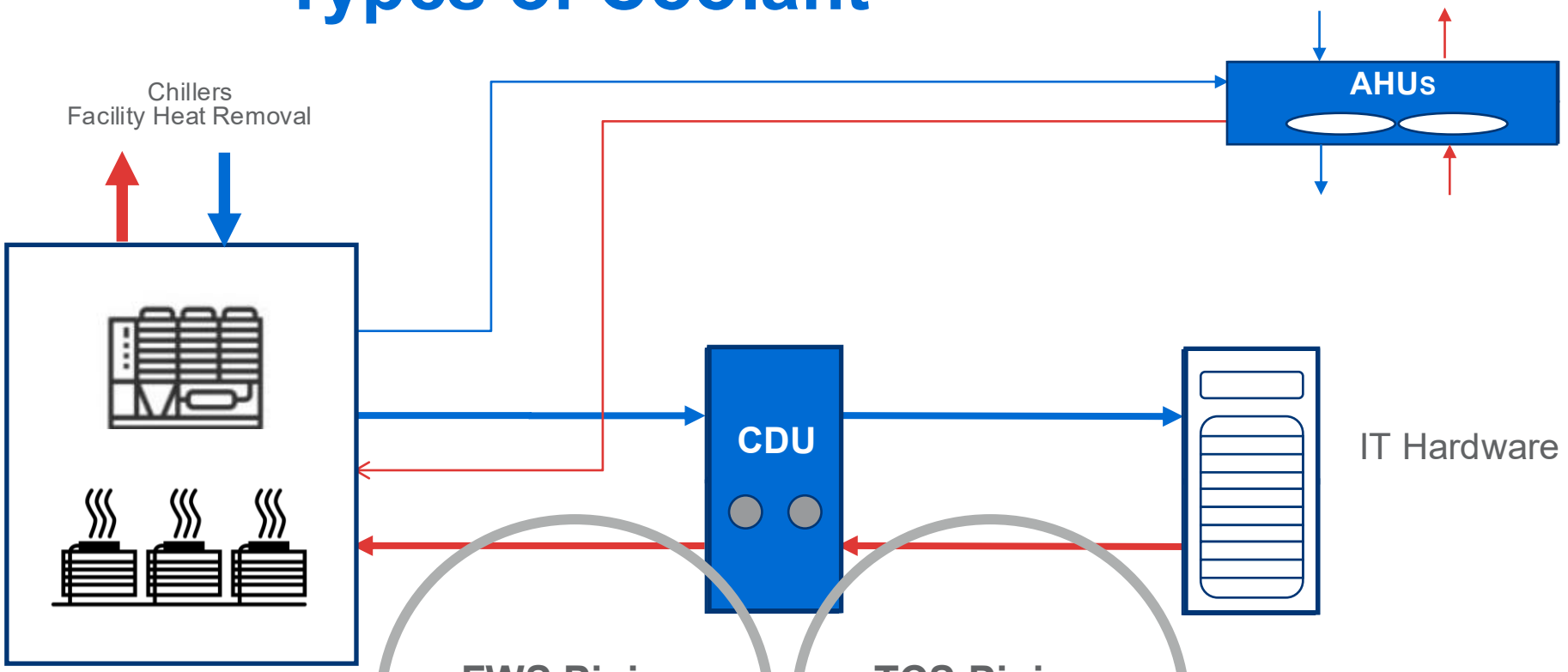
Water has better heat-transfer properties than PG25.

So why is the industry still running **glycol**?

The answer is a tradeoff you can measure. **Thermal. Operational. Programmatic.**

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Types of Coolant



FWS – Facility Loop
- 25–50% glycol – for freeze protection
- Corrosion inhibitors (azoles)

FWS Piping
Facility Loop

TCS Piping
Technology
Loop (D2C)

TCS – Technology Loop (D2C)
- PG25 glycol – for microbial protection, not freeze
- Treated water – needs added biocontrol

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Materials compatibility is a precondition. Wetted materials must be qualified against any chosen coolant.

Data Center Challenges with D2C Cooling Fluids

Unreliable supply of glycol coolants

Glycol supply inconsistencies pose risks in maintaining adequate cooling.

Thermal penalty of using glycol

Glycol-based cooling systems are less efficient at heat rejection compared to water, limiting cooling capacity and compute power.

Limited partners for last mile service

Data centers need reliable partners to supply high-quality glycol or ready-to-use water-based coolants in significant quantities.

High maintenance of water-based coolants

Water requires high maintenance to ensure quality through chemical treatments to prevent corrosion and bacterial growth, which can be complex.

Risk of bacterial growth

Bacterial growth can cause system downtime, thermal penalty, and increased maintenance costs, complicating data center operations.

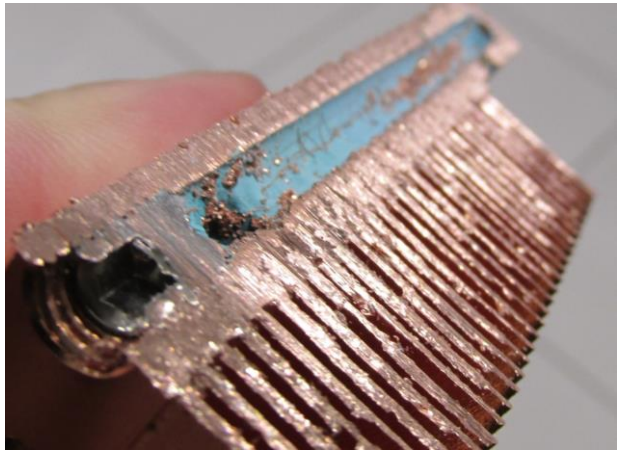
Sustainability & regulatory concerns

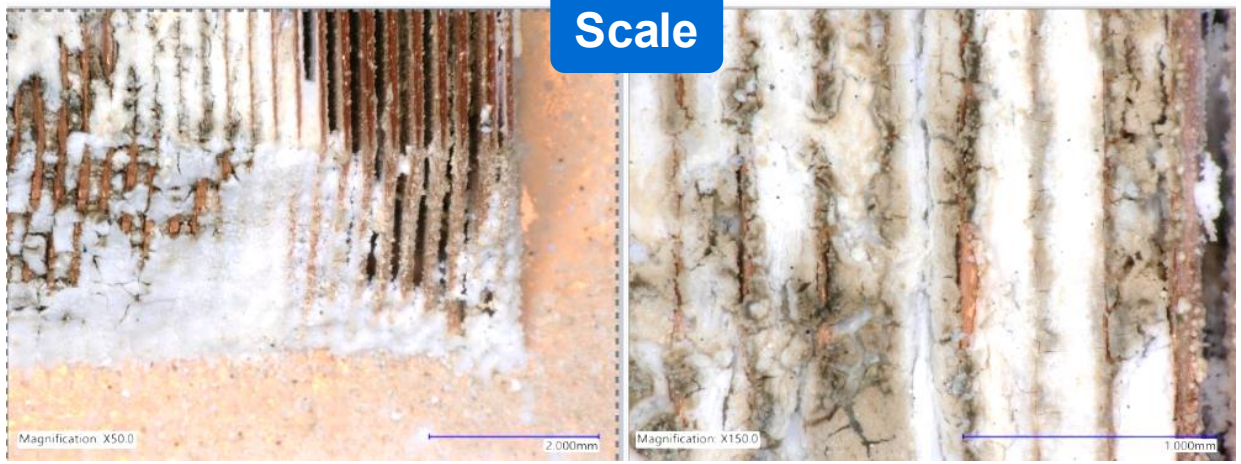
Glycols are typically derived from petroleum sources and bulk drain-discharge limits. Regulations require proper disposal of all azole-containing coolants at end-of-life.

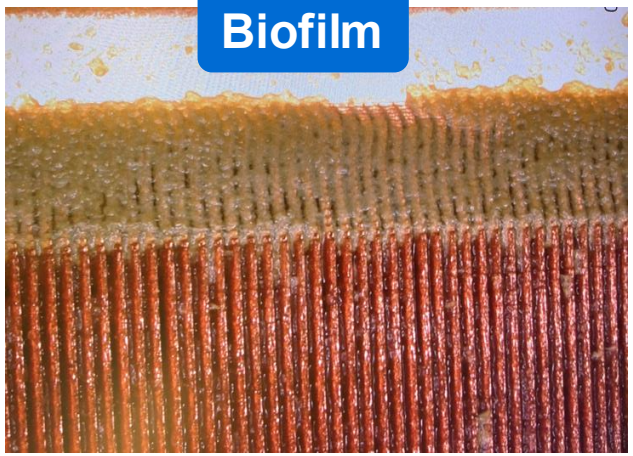
Glycol- and water-based systems each carry supply, performance, maintenance and sustainability trade-offs.

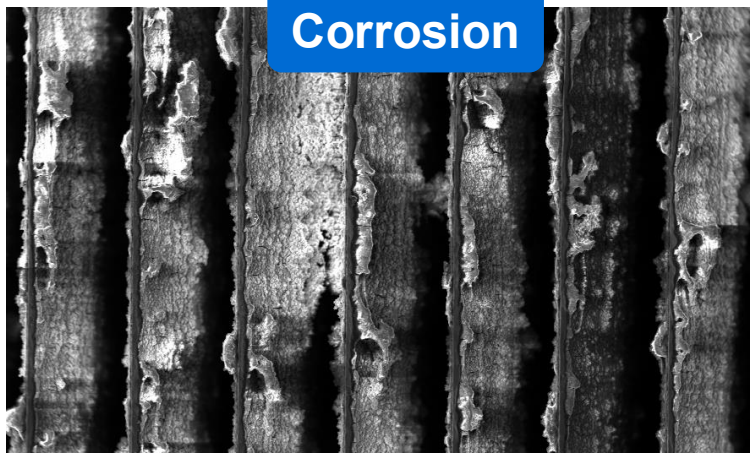
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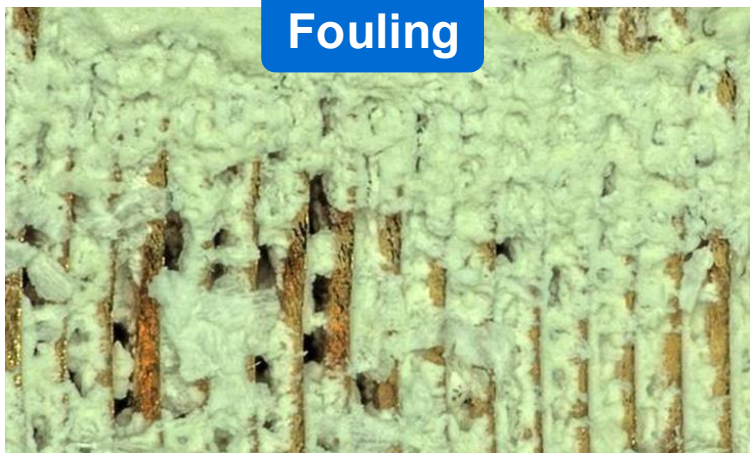
What Can Go Wrong in D2C Liquid Cooled Systems?











Each ended as downtime, lost cooling capacity, or cold plate replacement.

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Water & Glycol: Conventional Wisdom

Propylene Glycol (PG25)

Lower heat transfer capability

Lower maintenance

Higher energy cost

Treated Water

Better heat transfer properties

Higher maintenance

Lower energy cost

Choosing the Right Coolant for the Load

As density increases, the shift to water-based coolants could be crucial

PG25 predictable biocontrol | **Water** max cooling and energy efficiency

Property	PG25 Coolants	Water-based Coolants	What Does This Mean?
Thermal Conductivity (50°C)	Lower (~0.49 W/m·K)	Higher (~0.63 W/m·K)	Heat transfer penalty with propylene glycol
Specific Heat (50°C)	Lower (~3.9 kJ/kg·K)	Higher (~4.1 kJ/kg·K)	
Viscosity (50°C)	Higher (~1.3 mPa·s)	Lower (~0.5 mPa·s)	Increased energy cost to pump PG25 at higher viscosity
Pumping Power	Higher due to viscosity	Lower	
Biostatic Properties	Yes, if ≥ 24% PG	Requires pro-active steps to control bio-growth	Risk of biofilm is greater in water-based systems when not managed properly; PG25 is not risk-free
Maintenance Needs	Moderate	Higher (treat bio-growth)	Monitoring critical for any coolant; water-coolants typically require more intervention
Sustainability & Regulatory	Bulk glycol limits on drain discharge and azoles require regulated disposal	Smaller chemical footprint, but azoles still require regulated disposal	Both have azoles that require regulated disposal at end-of-life
Total Cost of Operations	Lower monitoring burden, higher fluid cost	Lower fluid cost, higher monitoring burden	TCO depends on site scale, labor model, and service access.

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Critical PG25 Operational Parameters

PARAMETER	ACCEPTABLE VALUES	WHY?
% PG	24 – 29% v/v	Biostatic protection
Conductivity	Trending	Can indicate over-dilution if decreasing, or contamination if increasing
Azole levels	> 400 ppm	Ensures corrosion protection
pH	8.0 – 10.5 (PBAs) 6.5 – 9.5 (OATs)	Ensures corrosion protection
Reserve alkalinity (phosphate buffer)	> 4 mL (PBAs) N/A (OATs)	Coolant's buffering capacity
Cu, Fe	< 3 ppm	Indicates corrosion or erosion
Ca, Mg	< 50 ppm	Can cause scaling; Indication of contamination
Cl, SO_4	< 25 ppm	Can cause corrosion; Indication of contamination
Turbidity (via IR)	Trending	Can indicate particulate contamination
Microbial counts (less frequent)	Stable counts over time	Can lead to biofilm growth; Can indicate problem if counts are increasing

Coolant Dependent

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*From *2026 Guidelines For Using Propylene Glycol-based Heat Transfer Fluids in Liquid Cooled Racks* and Ecolab experience

PG25 vs Water: Similar Parameters, Different Limits

Parameter	PG25 Coolants	Water Coolants	Why?
Microbial counts	Stable counts over time (keep PG > 24%)	Stable counts over time < 1000 CFU/mL ideal	Can lead to biofilm growth; Can indicate problem if counts are increasing
Azole levels	> 400 ppm	> 400 ppm	Ensures corrosion protection
Conductivity	Trending	Trending – may not be useful if low conductivity coolant	Can indicate over-dilution if decreasing, or contamination if increasing
pH	8.0 – 10.5 (PBAs) 6.5 – 9.5 (OATs) *Coolant Dependent*	8 – 10.5 (typically)	Ensures corrosion protection
Reserve alkalinity	> 4 mL (PBAs) N/A (OATs)	May not be needed if not buffered *Coolant Dependent*	Coolant's buffering capacity
Cu, Fe	< 3 ppm	< 0.1 ppm (may be higher if introduced from biocides)	Indicates corrosion or erosion
Ca, Mg	< 50 ppm	< 30 ppm (may have higher/lower tolerance dependent on coolant ionic strength)	Can cause scaling; Indication of contamination
Cl, SO_4	< 25 ppm	< 25 ppm	Can cause corrosion; Indication of contamination
Turbidity	Trending	Trending	Can indicate particulate contamination

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See OCP's *Guidelines for Using Water-Based Transfer Fluids in Single-Phase Cold Plate-Based Liquid-Cooled Racks*

How to leverage the thermal benefits of water?

Density is still climbing: the question is not IF water, but HOW to run it

- Water leads on thermal conductivity and heat capacity
- The thermal advantage only holds in clean, protected loops
- Maintaining PG concentration (>24%) supports biostatic protection
- Water-based coolants need direct biocontrol intervention
- Maintenance is a key factor in the TCO trade-off. Closing the maintenance gap is critical for leveraging benefits of water

Clean Startup

Guided cleaning, commissioning and frequent early sampling, so the loop starts clean and heat transfer begins at design.

Engineered Chemistry

Corrosion protection plus a defined biocontrol package, so water's thermal advantage is never traded for metal loss or fouling.

Operational Fluid Assurance

Field testing, lab analyses, trend interpretation and troubleshooting that hold microbial and azole levels in range over time.

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Ecolab's Approach to Coolants

Programmatic approach is what converts water's thermal advantage into sustained performance

All Coolants

- End-to-end services providing faster commissioning and start-up
- Analytical support combined with 3D TRASAR™ technology, providing continuous, real-time monitoring
- Address remediation as needed during startup and operations

Glycol Coolants

- Strengthening supply reliability through diversified, responsible sourcing
- Helping maintain glycol quality during operations to preserve biostatic and corrosion protection

Water-Based Coolants

- Developing coolants that require less intervention, closing the maintenance gap with PG25
- Hypercare startup/commissioning, biocide/chemical dosing guidance and fluid assurance during operations

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A Practical Decision Framework

Limited on-site chemistry and service capability	PG25
Rack energy density needs maximum cooling	WATER
Monitoring program and service partner in place	WATER
Freeze exposure in makeup path	PG25

Multiple factors compete. The right fluid is the one that balances them for your operation.

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Let’s compare notes!

Visit Ecolab in Hall 3 at booth E19 & E9

01 Who here is running glycol, who is running water, and who has run both?

02 What worries you most about your coolant?

03 What has caused your biggest downtime event?

04 Where does your coolant spec come from?



Thank you for your time and attention!

Dolly Batra, Ph.D., Sr. RDE Group Leader

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Appendix

How Does Coolant Failure Happen?

- Contamination or dilution with node changeouts or coolant top-off
- Design flaw / wetted material incompatibility
- Critical component malfunction
- Use of incompatible coolants during top-off or coolant changeover
- Coolant leaks

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PG25 Products Landscape

Product Specifications	Phosphate-Based Azoles (PBAs)	Organic Acid Technologies (OATs)
Propylene glycol 25% v/v	✓	✓
Copper corrosion inhibitor (azoles)	✓	✓
pH	8 – 10.5	6.5 – 9.5
Reserve alkalinity (buffering capacity)	✓	Minimal by design
Low heavy metals	✓	✓
Other inhibitors	None	Organic acids

Critical Criteria

- ✓ High purity to prevent scaling, clogging or corrosion
- ✓ Sufficient azole content to prevent copper cold plate corrosion
- ✓ NVIDIA DSX Ready Coolant

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Typical Water-Based Coolants

Ready-to-use water-based coolants or mixed on-site with high-purity water

More varying chemistries and characteristics than PG25

Critical components:

- Copper corrosion inhibitors (Azoles)
- Protection against microbial growth

See OCP's *Guidelines for Using Water-Based Transfer Fluids in Single-Phase Cold Plate-Based Liquid-Cooled Racks*

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